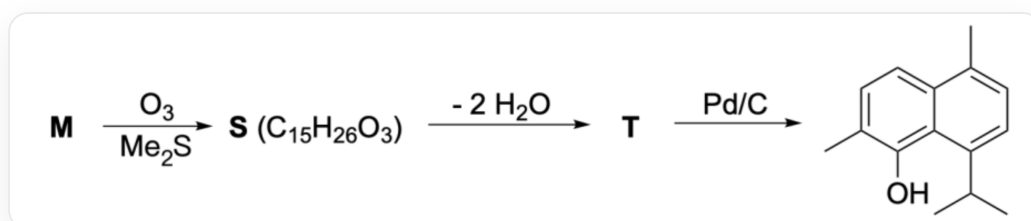


Question

Ozonolysis is often used for the determination of organic structures. Compound **M** ($C_{15}H_{26}O$) is an alcohol that is difficult to hydrogenate catalytically. When **M** is treated with sulfur, an aromatic compound **R** ($C_{15}H_{18}$) is ultimately obtained. Ozonolysis of **M** followed by additional treatments yields a naphthalene derivative, as shown in the following reaction:



The reaction is represented by SMILES: [M]>[O-][O+]=O.CSC>[S]>>

[T]>Pd/C>CC(C)C1=CC=C(C)C2=CC=C(C)C(=C21)O, where [S] and [T] are compound designations rather than elemental symbols. The chemical formula of [S] is $C_{15}H_{26}O_3$, and the conversion from [S] to [T] involves the elimination of two molecules of water

It is known that the hydroxyl group in **M** is located on a tertiary carbon outside the ring. Based on this, determine which of the following options are correct:

- M** has 3 chiral carbons.
- M** has 4 chiral carbons.
- M** contains two six-membered rings, and the double bond is in the cis configuration.
- M** contains two six-membered rings, and the double bond is in the trans configuration.
- M** contains one five-membered ring and one seven-membered ring, and the double bond is in the cis configuration.
- M** contains one five-membered ring and one seven-membered ring, and the double bond is in the trans configuration.
- M** contains one six-membered ring and two double bonds.

8. In **R**, a maximum of 12 carbons lie in the same plane.
9. In **R**, a maximum of 13 carbons lie in the same plane.
10. The conversion from **T** to the product involves the elimination of 1 molecule of H_2 .
11. The conversion from **T** to the product involves the elimination of 2 molecules of H_2 .
12. The conversion from **T** to the product involves the elimination of 3 molecules of H_2 .

A. 1, 3, 8, 10

B. 1, 4, 9, 11

C. 1, 7, 10

D. 1, 6, 11

E. 1, 5, 9

F. 1, 6, 9, 11

G. 1, 4, 8, 11

H. 1, 7, 12

I. 2, 6, 9, 11

J. 2, 6, 11

K. 2, 5, 12

L. 2, 3, 8, 10

M. 2, 7, 12

N. 3, 9, 10

O. 6, 8, 11

P. 7, 11

Q. 5, 12

R. None of the above options is correct

Answer

Correct Answer: **D**

Detailed Explanation

M is converted to **S** via ozonolysis with unchanged carbon count, indicating the cleavage of an internal double bond in the ring.

CHECKPOINT

1 PTS

The cleavage from **M** to **S** involves an internal double bond in the ring

Ozonolysis does not alter hydroxyl groups, so **S** still retains a hydroxyl group. It can be inferred that the two dehydration steps from **S** to **T** consist of one aldol condensation + elimination and one elimination reaction.

CHECKPOINT

1 PTS

The two dehydration steps from **S** to **T** consist of one aldol condensation + elimination and one elimination reaction

From the chemical formula of **S**, it is known that **T** has the formula $C_{15}H_{22}O$, while the product formula is $C_{15}H_{18}O$, indicating the loss of 2 molecules of H_2 . Thus, option 11 is correct, while options 10 and 12 are incorrect.

CHECKPOINT

1 PTS

The conversion from **T** to the product involves the loss of 2 molecules of H_2

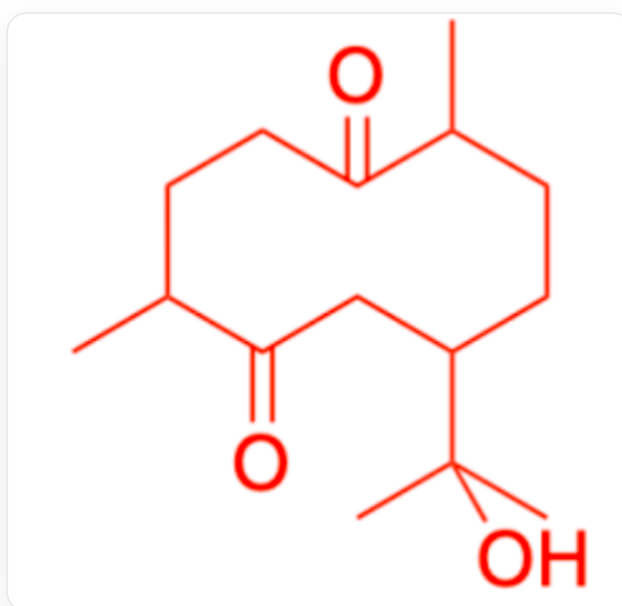
It is evident that the ring skeleton of **T** remains unchanged in the product. According to the question, the hydroxyl group in **M** is located on a tertiary carbon outside the ring. Therefore, the hydroxyl group in **M** and **S** can only reside on the carbon corresponding to the sole tertiary carbon in the product.

CHECKPOINT

1 PTS

The hydroxyl group in **M** and **S** can only reside on the carbon corresponding to the sole tertiary carbon in the product

The product is a fused ring system formed via aldol condensation. The shared C-C bond between the two rings can be cleaved, and the carbonyl oxygen is connected to the end farther from the -OH group. Based on the chemical formula and degree of unsaturation of **S**, its structure is determined as:

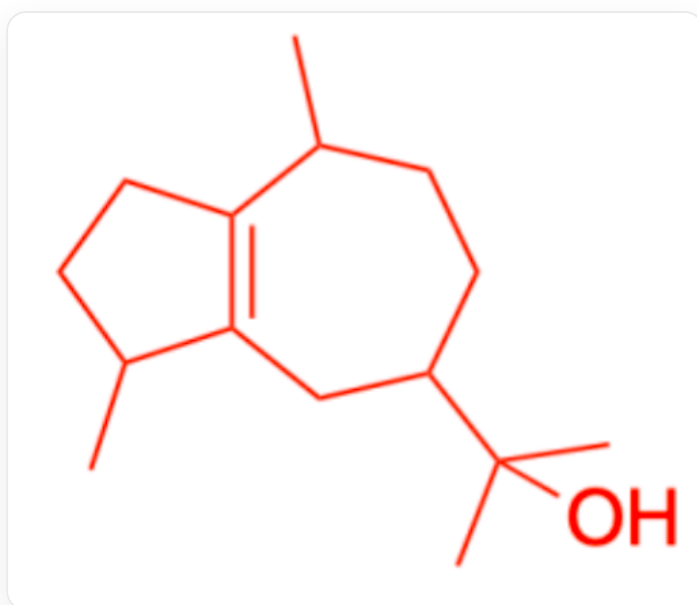

CC1CCC(CC(=O)C(C)CCC1=O)C(C)(C)O

CHECKPOINT

1 PTS

S is CC1CCC(CC(=O)C(C)CCC1=O)C(C)(C)O

Ozonolysis cleaves double bonds into two carbonyl groups, so the structure of **M** can be deduced from **S**:



CC1CCC(CC2=C1CCC2C)C(C)(C)O

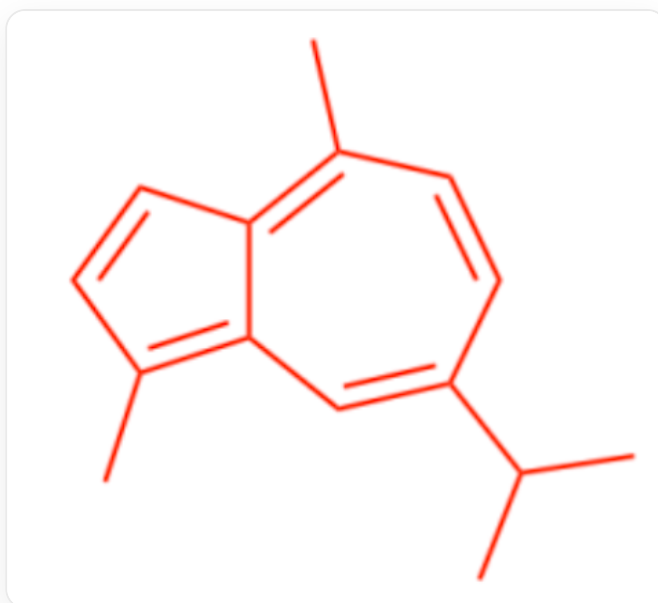
CHECKPOINT

1 PTS

M is CC1CCC(CC2=C1CCC2C)C(C)(C)O

The difficulty in catalytic hydrogenation is due to the double bond in **M** being tetrasubstituted, with bulky groups hindering its interaction with the catalyst surface.

Evidently, after sulfur treatment, **M** transforms into the aromatic **R**:



CC(C)C1=CC2=C(C)C=CC2=C(C)C=C1

CHECKPOINT

1 PTS

R is CC(C)C1=CC2=C(C)C=CC2=C(C)C=C1

The correctness of the options can now be determined:

From **M** being CC1CCC(CC2=C1CCC2C)C(C)(C)O, it is clear that **M** has 3 chiral carbons, making option 1 correct and 2 incorrect. **M** contains one five-membered ring and one seven-membered ring, with the double bond in a trans configuration, so option 6 is correct, while options 3, 4, 5, and 7 are incorrect.

CHECKPOINT

1 PTS

M has 3 chiral carbons

CHECKPOINT

1 PTS

M contains one five-membered ring and one seven-membered ring, with the double bond in a trans configuration

From **R** being CC(C)C1=CC2=C(C)C=CC2=C(C)C=C1 and the tetrahedral geometry of sp^3 carbons, it follows that a maximum of 14 carbons in **R** can lie in the same plane. Thus, options 8 and 9 are incorrect.

CHECKPOINT

1 PTS

A maximum of 14 carbons in **R** can lie in the same plane

In summary, options 1, 6, and 11 are correct, and the answer is D.